

Question

Diverse novel species containing peroxo bonds are prepared using different methods.

Compound **A** can be prepared by bubbling CO_2 into a concentrated aqueous hydrogen peroxide solution containing KOH , crystallizing, while controlling the temperature at -5°C to -20°C .

Compound **B** can be prepared by adding potassium bicarbonate to saturation in a 30% aqueous hydrogen peroxide solution, then adding a certain amount of ethanol, and cooling to crystallize, while controlling the temperature at -10°C to -15°C .

Crystal structure determination and thermal analysis indicate that:

In the crystal of **A**, there is only one type of anion;

In the crystal of **B**, there is only one type of cation, one type of anion, and one type of neutral molecule.

A loses 30.3% of its weight when heated above 160°C (Reaction 1);

B loses 53.9% of its weight when heated above 200°C (Reaction 2).

The following statements are made:

1. The anion structure of **A** contains a mirror plane.
2. The ratio of the number of cations, anions, and neutral molecules in the composition of **B** is 1 : 1 : 1.
3. The solid products of the thermogravimetric analysis of **A**, **B** are consistent.
4. The coefficient of H_2O in the equation of Reaction 2 (with coprime integer coefficients) is 3. (If there is no H_2O in the reaction equation, this statement is considered correct)
5. Arranging **A**, K_2O_2 , O_2F_2 according to the length of the peroxo bond, the order is $\text{K}_2\text{O}_2 > \text{O}_2\text{F}_2 > \text{A}$.

What is the sum of the serial numbers of the correct statements?

A. 2

B. 3

C. 5

D. 6

E. 8

F. 9

G. 11

H. 12

I. 14

J. All of the above options are incorrect.

Answer

Correct Answer: **F**

Detailed Explanation

First consider the final product of thermogravimetric analysis. KHCO_3 decomposes at 120°C , so the final solid product of thermogravimetric analysis is K_2CO_3 . The solid products of thermogravimetric analysis of **A**, **B** are consistent, so statement 3 is correct.

CHECKPOINT

1 PTS

The final solid products of thermogravimetric analysis of **A**, **B** are both K_2CO_3

Assume **A** has 2 potassium atoms, then its molecular weight is $138.31 \div (1 - 0.303) = 198.44$. Combining the preparation conditions of strong alkalinity, we know that **A** is $\text{K}_2\text{C}_2\text{O}_6$.

CHECKPOINT

1 PTS

A is $\text{K}_2\text{C}_2\text{O}_6$

Its anion structure is two $[\text{CO}_3^-]$ groups connected by a peroxo bond, similar to the structure of H_2O_2 , with a two-fold rotation axis, but no mirror plane or center of symmetry, so statement 1 is incorrect.

CHECKPOINT

1 PTS

The anion structure of **A** does not have a mirror plane

Similarly, assume **B** has 2 potassium atoms, then its molecular weight is $138.31 \div (1 - 0.539) = 300.02$. Its cation is $2K^+$, and according to the preparation process, it is inferred that the anion is $2HOOCO_2^-$, and the remaining molecular weight of about 68 is $2H_2O_2$, which is consistent with the description of the question, so **B** is $KHCO_4 \cdot H_2O_2$. Other compositions cannot satisfy the thermogravimetric data. The ratio of the number of cations, anions, and neutral molecules in the composition of **B** is 1 : 1 : 1, so statement 2 is correct.

CHECKPOINT

2 PTS

B is $KHCO_4 \cdot H_2O_2$

Reaction 2 is the decomposition of $KHCO_4 \cdot H_2O_2$, that is $2[KHCO_4 \cdot H_2O_2] \rightarrow K_2CO_3 + CO_2 + 3H_2O + 2O_2$. The coefficient of H_2O is 3, so statement 4 is correct.

CHECKPOINT

2 PTS



The negative charges on the two oxygen atoms in O_2^{2-} have repulsive force, so the peroxo bond is the longest; in O_2F_2 , the lone pair electrons of oxygen are hyperconjugated with the antibonding orbital of the other oxygen-fluorine bond, which enhances the bond order of the peroxo bond, so the peroxo bond is the shortest; the hyperconjugation effect brought by the CO_2^- group in **A** is not as good as O_2F_2 . Therefore, the order of the peroxo bond length is $K_2O_2 > \mathbf{A} > O_2F_2$, so statement 5 is incorrect.

CHECKPOINT

1 PTS

In O_2F_2 , the lone pair electrons of oxygen are hyperconjugated with the antibonding orbital of the other oxygen-fluorine bond, enhancing the bond order of the peroxy bond

CHECKPOINT

1 PTS

Peroxy bond length order $\text{K}_2\text{O}_2 > \mathbf{A} > \text{O}_2\text{F}_2$

The correct statements are 2, 3, and 4, so the answer is option F.